



Technical Service Bulletin

Technical Service Bulletin: TSB210148

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Cooling Fan Drive Belt and Accessory Drive Belt Walking, Shredding, and or Falling Off

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

Product Affected

- QST30 CM552
- QST30 CM2350 T101
- QST30 CM2350 T103C

Issue

Symptoms:

- Engine experiences cooling fan drive belt coming off fan hub pulley and or shredding within five months in service.
- Engine derate
- Coolant Temperature Above Normal - Gradual Overheat
- Coolant Temperature Above Normal - Sudden Overheat

Root Cause:

- Linear tensioner rod ends could be broken.
- Idler lever pulley could have excessive play in the pulley.
- Scenarios by malfunction type:
 - Scenario 1: Drive belt shreds because of misalignment with idler lever pulley
 - Scenario 2: Drive belt walks off crankshaft pulley because of misalignment and or tension force **not** to specification
 - Scenario 3: Drive belt walks off crankshaft pulley due to fan blade inertia /weight/size change. Drive belt shredding or coming off crankshaft pulley occurs ten or less hours or within first six months in operation.

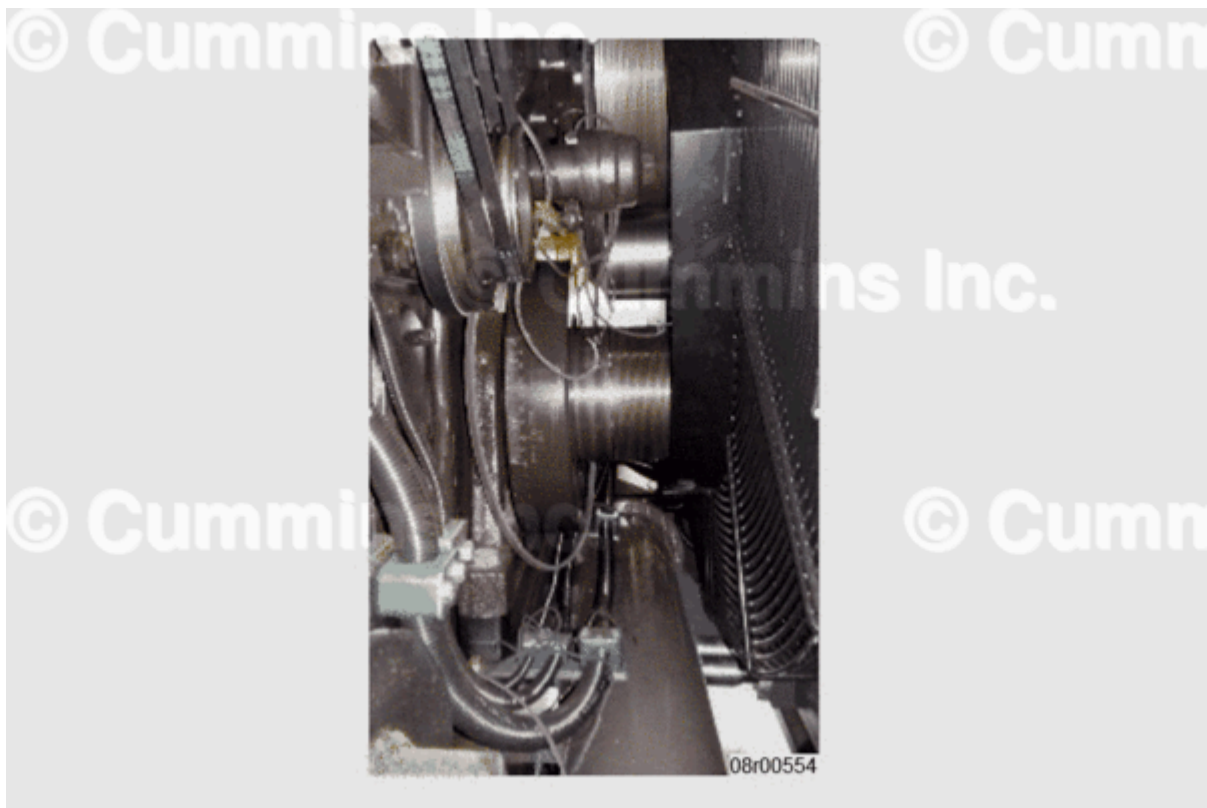


Figure 1, Example of Belt Shredding.

Verification

If any of the symptoms above are detected, investigate using procedures as stated in Publications Table 3 below.

Malfunction Verification:

- Confirm drive belt shreds and or walks off vibration damper adapter pulley
- Check linear tensioner rod ends for breakage
- Check alignment of all pulleys with each other to verify all of them are on same plane.
- Follow QuickServe® Online (QSOL) for alignment procedures.
- If alignment is off and idler lever pulley has excessive play:
 - Remove idler lever pulley assembly and disassemble.
 - Check for excessive sleeve wear or seizure.
- If alignment is within specification:
 - Check belt tension force (belt is intact) or replace and set tension force. Follow QSOL procedures for checking alignment, troubleshooting fault codes, and inspecting part wear.
- If all components are within specification, verify fan blade assembly meets requirements according to procedures on QSOL.

Resolution

New service parts solutions have been implemented to lessen or resolve the issues stated above. See Service Instructions section below for specific direction based on different scenarios.

- New tensioner assembly has new design rod ends installed
- New tensioner rod ends have a stronger material that has a higher yield strength than existing design

- Rod ends for production intent material change are approved parts **not** made on production tooling whereas production permanent material change parts are manufactured on production tooling. Both tooling material changes are better than old rod end material.
- Once production intent rod end inventory depletes to zero, production permanent parts will be the **only** rod end for engine platform
- New idler lever assembly has new design sleeves installed.
- New idler lever sleeves are a change from nylon sleeve to oil impregnated bronze sleeves.
 - New assembly will require four sleeves for better stability and robustness of pivot joint.

Service Instructions

Scenario 1:

- If idler lever pulley sleeves have excessive wear and or have seized condition (see Figure 2 below), replace sleeves with four new bronze oil impregnated sleeves.
- Part numbers provide in Service Parts Availability Table 1 below.

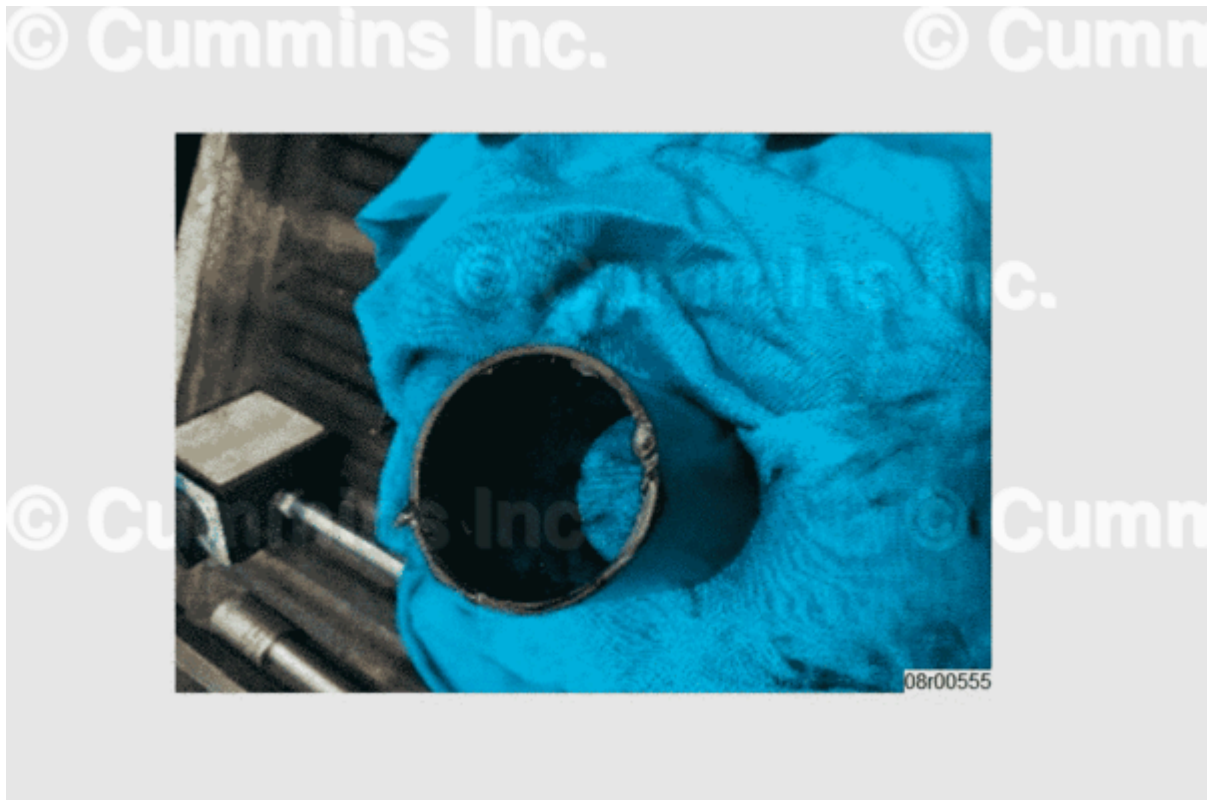


Figure 2, Nylon Sleeve Edge Loaded.

Scenario 2:

- If fan drive belt linear tensioner rod ends are broken (see Figure 3 below), replace tensioner assembly or just replace rod ends as provided in Service Parts Availability Table 1 below.



Figure 3, Male Rear Rod End Broken at Shaft.

Scenario 3:

- If all items check good for alignment, belt tension, rod ends in good condition, and fan blade requirements are met, replace the damper pulley adapter with a double lip pulley in comparison to the single lip pulley.
- Extra lip prevents belt from walking off crankshaft pulley. See Technical Service Bulletin, Cooling Fan Drive Belt Walking or Shredding, TSB210084 for part number information.

Service Parts Availability

Service parts are available. See Table 1 for part numbers.

Table 1, Service Parts				
Part Description	Existing Part Number	Obsolete	Superseded	New/Alternative Part Number
End, Control Rod (Male thread)	3347591	No	No	5593601
End, Control Rod (Female thread)	177269	No	No	5593603
Tensioner, Belt	2888890	No	No	5593585
End, Control Rod (Male thread)	3347591	No	Yes	5593573
End, Control Rod (Female thread)	177269	No	Yes	5593574

Table 1, Service Parts				
Part Description	Existing Part Number	Obsolete	Superseded	New/Alternative Part Number
Assembly, Fan Drive Idler (New sleeves)	4975421	No	No	Same
Sleeve	3009477	No	Yes	5593572

Part Inventory Action

- If customer and or distributor order requests for new production intent part number, release and grant order.
- If the customer and or distributor order requests for existing part number, continue to honor those orders.
- Future production permanent rod ends will supersede current and old rod end once available inventory is zero.

Production Status

Implemented for production. See Table 2.

Table 2, Production Information			
ESN First	Build Date ¹	Plant	Part Number Change
37284240	26-March-2021	Seymour Engine Plant	5593601
37284240	26-March-2021	Seymour Engine Plant	5593606
37284951	2-July-2021	Seymour Engine Plant	5593585
37284951	2-July-2021	Seymour Engine Plant	5593573
37284951	2-July-2021	Seymour Engine Plant	5593574
37283660	5-March-2021	Seymour Engine Plant	5593601
37283660	5-March-2021	Seymour Engine Plant	5593606
¹ Engine build date can be found on the engine dataplate.			

Publications Affected

Table 3, Publications Affected					
Manual Type	Engine	Bulletin Number	Procedure Title	Procedure	Section
Service Manual	QST30 CM552	4021539	Cooling System- Drive Belt, Cooling Fan	Refer to Procedure 008-002 (/qs3/pubsy s2/xml/en/pr ocedures/57 /57-008-002-tr.html)	8
Service Manual	QST30 CM552	4021539	Cooling System- Fan Drive Idler Arm Assembly	Refer to Procedure 008-029 (/qs3/pubsy s2/xml/en/pr ocedures/57 /57-008-029.html)	8
Service Manual	QST30 CM552	4021539	Cooling System- Fan, Cooling	Refer to Procedure 008-040 (/qs3/pubsy s2/xml/en/pr ocedures/57 /57-008-040-tr.html)	8
Service Manual	QST30 CM552	4021539	Cooling System- Cooling Fan Belt Tensioner	Refer to Procedure 008-087 (/qs3/pubsy s2/xml/en/pr ocedures/57 /57-008-087-tr.html)	8
Service Manual	QST30 CM2350	5467179	Cooling System- Drive Belt, Cooling Fan	Refer to Procedure 008-002 (/qs3/pubsy s2/xml/en/pr ocedures/36 3/363-008-002-tr.html)	8

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Manual Type	Engine	Bulletin Number	Procedure Title	Procedure	Section
Service Manual	QST30 CM2350	5467179	Cooling System- Fan Drive Idler Arm Assembly	Refer to Procedure 008-029 (/qs3/pubsy s2/xml/en/pr ocedures/36 3/363-008-029-tr.html)	8
Service Manual	QST30 CM2350	5467179	Cooling System- Fan, Cooling	Refer to Procedure 008-040 (/qs3/pubsy s2/xml/en/pr ocedures/36 3/363-008-040.html)	8
Service Manual	QST30 CM2350	5467179	Cooling System- Cooling Fan Belt Tensioner	Refer to Procedure 008-087 (/qs3/pubsy s2/xml/en/pr ocedures/36 3/363-008-087.html)	8

Document History

Date	Details
2021-7-8	Module Created
2021-9-22	Updated Table 1 with part number 5593585.